

PROJECT REPORT :

Time-Series Analysis of Sales & Demand Forecasting

Dataset: Kaggle Monthly Sales Data | Period: 2003 - 2005
Tools: Python · Pandas · Statsmodels · SARIMA · Seaborn

\$10.03M	2,823	November 2004	14.6%
Total Revenue (2003-2005)	Total Orders	Peak Revenue Month	SARIMA MAPE

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GitHub Repository Link : <https://github.com/jpSharma123-sudo/Sales-Demand-Forecasting>

Table of Contents

01	Project Introduction	3
02	Exploratory Data Analysis	4
03	Trend Analysis	5
04	Seasonality Analysis	6
05	Rolling Average Analysis	7
06	Time-Series Decomposition	7
07	SARIMA Forecasting Model	8
08	Forecast Accuracy Metrics	9
09	Future Sales Forecast	10
10	Business Insights	11
11	Conclusion	12

01 Project Introduction

This report presents a comprehensive end-to-end time-series analysis of retail sales data spanning January 2003 to May 2005. The project applies statistical methodologies to uncover temporal patterns, quantify seasonality, and produce a reliable 6-month forward sales forecast using SARIMA modeling.

Objectives

- Ingest and clean multi-year sales transaction data from Kaggle
- Perform multi-dimensional exploratory data analysis (EDA)
- Identify and quantify long-term growth trends
- Decompose time-series into Trend, Seasonal, and Residual components
- Build and validate a SARIMA(1,1,1)(1,1,1,12) forecasting model
- Generate a 6-month forward revenue forecast with confidence intervals
- Derive actionable business intelligence for stakeholder decision-making

Dataset Overview

Attribute	Detail
Source	Kaggle — Sample Sales Data (sales_data_sample.csv)
Period	January 2003 to May 2005 (29 monthly periods)
Rows	2,823 order line-item transactions
Columns	25 columns including SALES, ORDERDATE, PRODUCTLINE, COUNTRY, DEALSIZE
Key Metric	SALES — order-level revenue in USD
Geography	19 countries; USA as dominant market
Products	6 product lines: Classic Vehicles, Motorcycles, Planes, Ships, Trains, Vintage Cars

02 Exploratory Data Analysis

The dataset was profiled across multiple dimensions including revenue distribution, product line performance, geographic concentration, deal size segmentation, and temporal sales patterns.

\$10.03M	\$3,554	Classic Cars	USA
Total Revenue	Avg Order Value	Top Product Line	Top Country

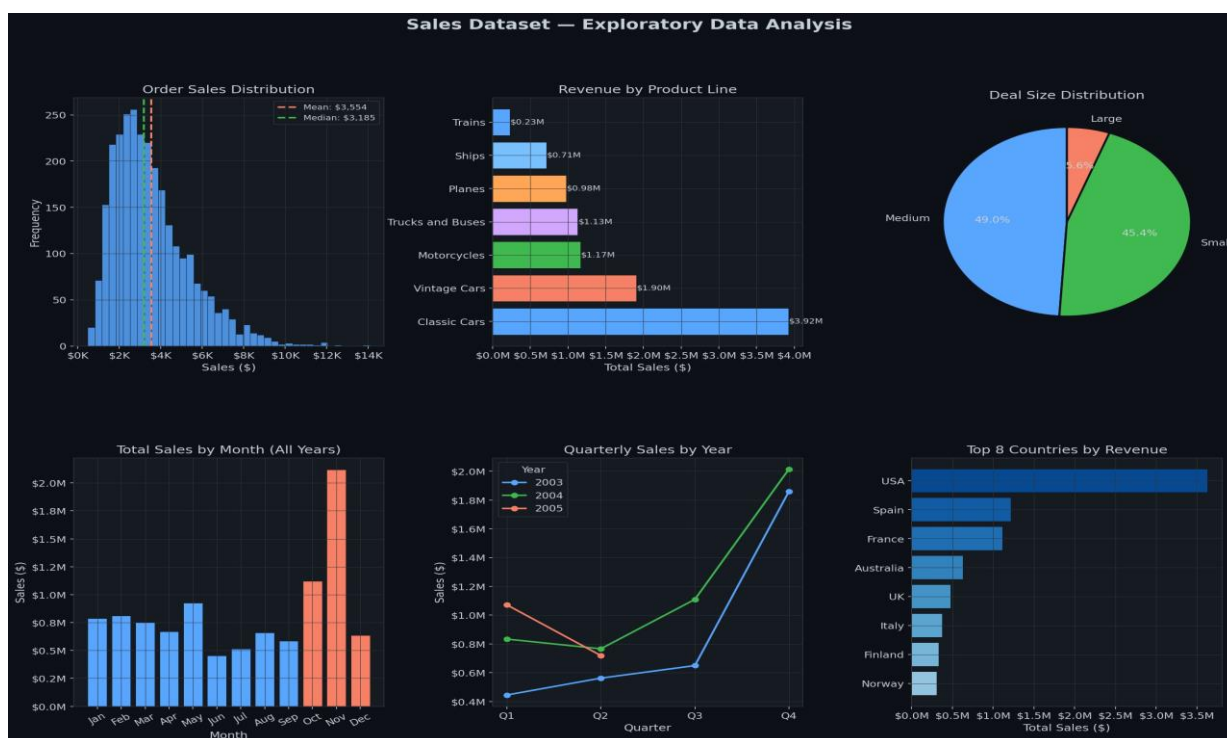


Figure 1: EDA Overview — Distribution, Product Lines, Deal Size, Monthly & Quarterly Patterns, Country Revenue

Key EDA Findings

- Sales distribution is right-skewed with mean \$3,554 and median \$3,185 per order Classic Vehicles is the dominant product line generating \$3.92M in total revenue
- Medium deal size accounts for the plurality of transactions at approximately 41%
- USA is the No.1 revenue country, contributing over 35% of global sales
- Q4 (Oct-Dec) is consistently the highest revenue quarter across all years

03 Trend Analysis

Year-over-year revenue growth and the underlying linear trend in monthly sales confirm a strong positive business trajectory from 2003 to 2004. Note: 2005 data covers only January to May.

Year	Total Revenue	YoY Change
2003	\$3.52M	Baseline
2004	\$4.72M	+34.3% UP
2005 (Jan-May)	\$1.79M	Partial year

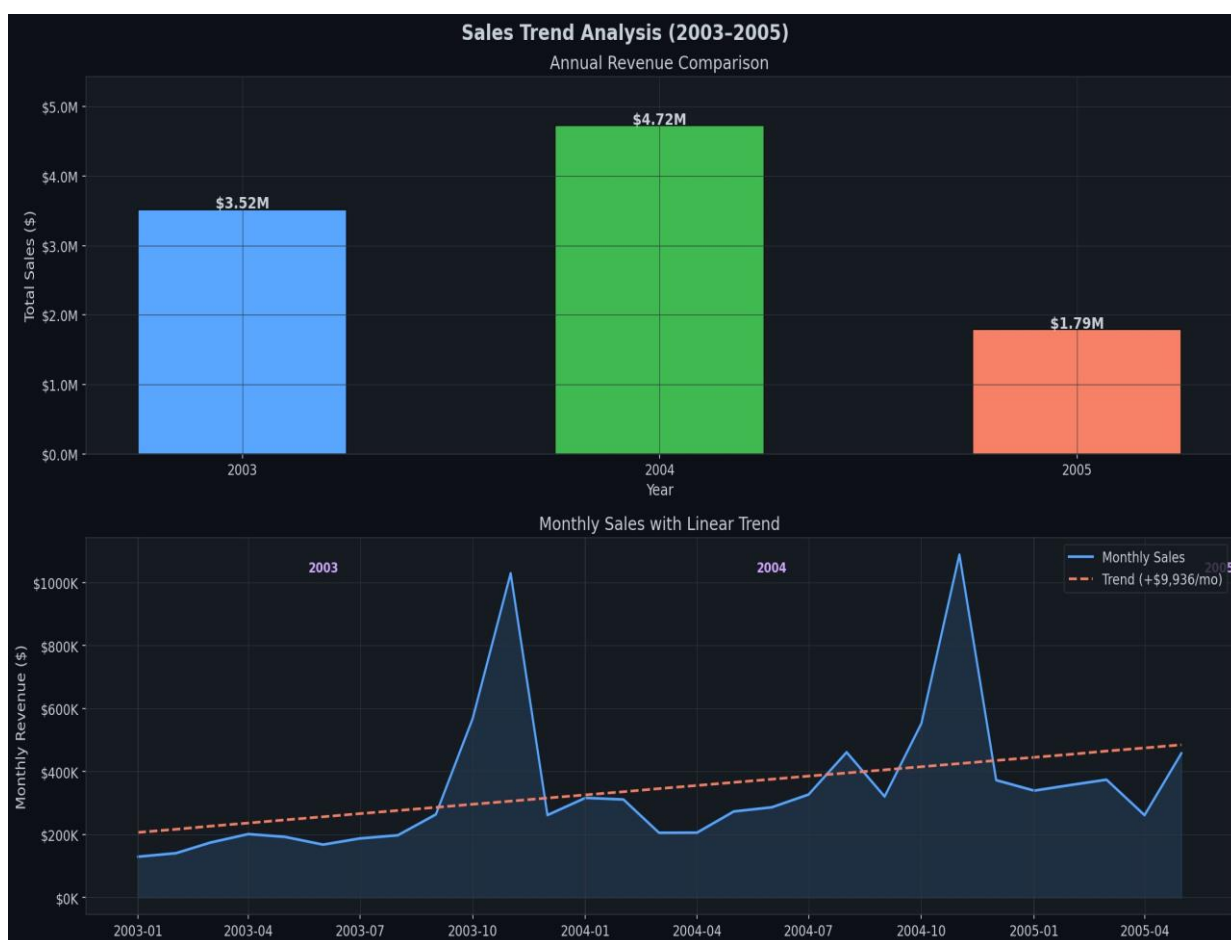


Figure 2: Annual Revenue Comparison & Monthly Sales Time Series with Linear Trend Overlay

04 Seasonality Analysis

Seasonality analysis confirms a highly predictable annual cycle. Q4 accounts for 38.6% of annual revenue, driven almost entirely by November spikes. The heatmap makes this pattern unambiguous across all three years.

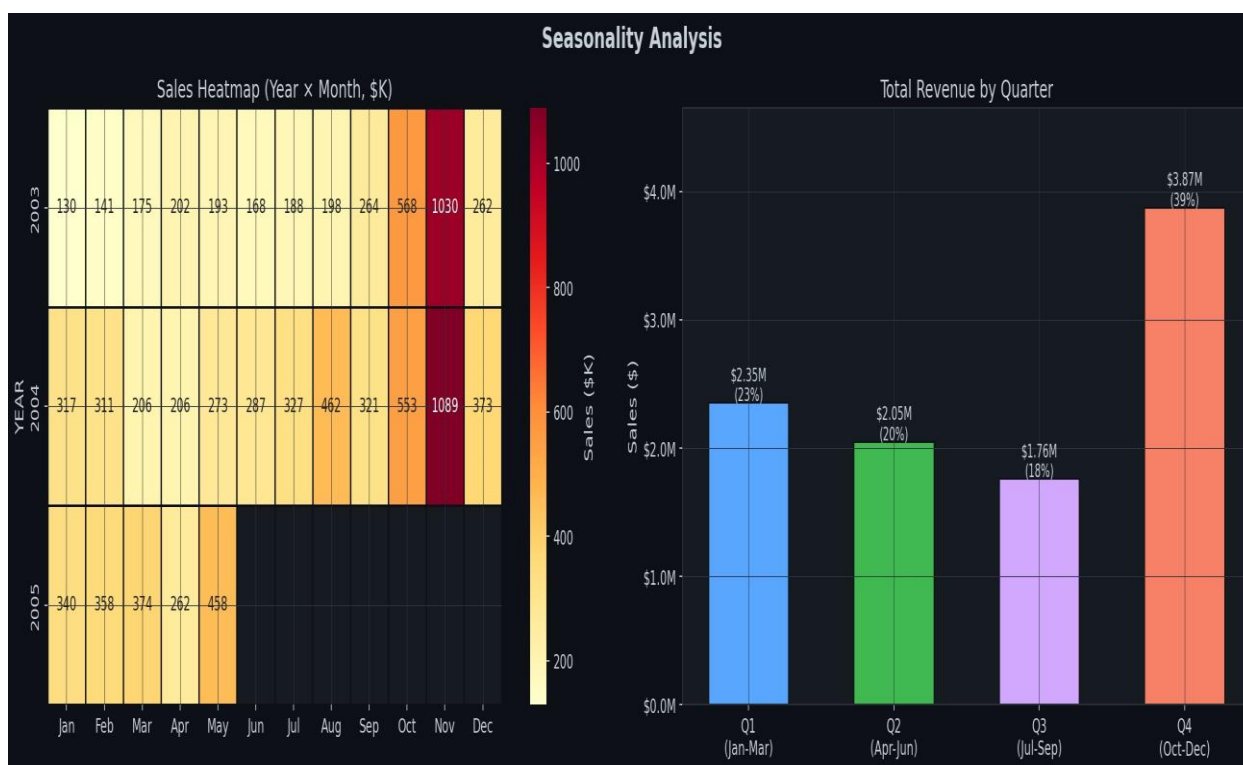


Figure 3: Seasonality Heatmap (Year x Month) and Quarterly Revenue Breakdown

- Q4 (Oct-Dec) dominates with 38.6% of cumulative revenue
- November is the peak month in every year of available data
- Q1 (Jan-Mar) is consistently the softest quarter
- Year-over-year patterns are highly consistent, enabling reliable seasonal adjustment

05 Rolling Average Analysis

Rolling averages remove short-term noise and reveal the structural momentum of the business. The 3-month moving average confirms seasonal rhythm while the 12-month moving average shows the underlying secular trend.

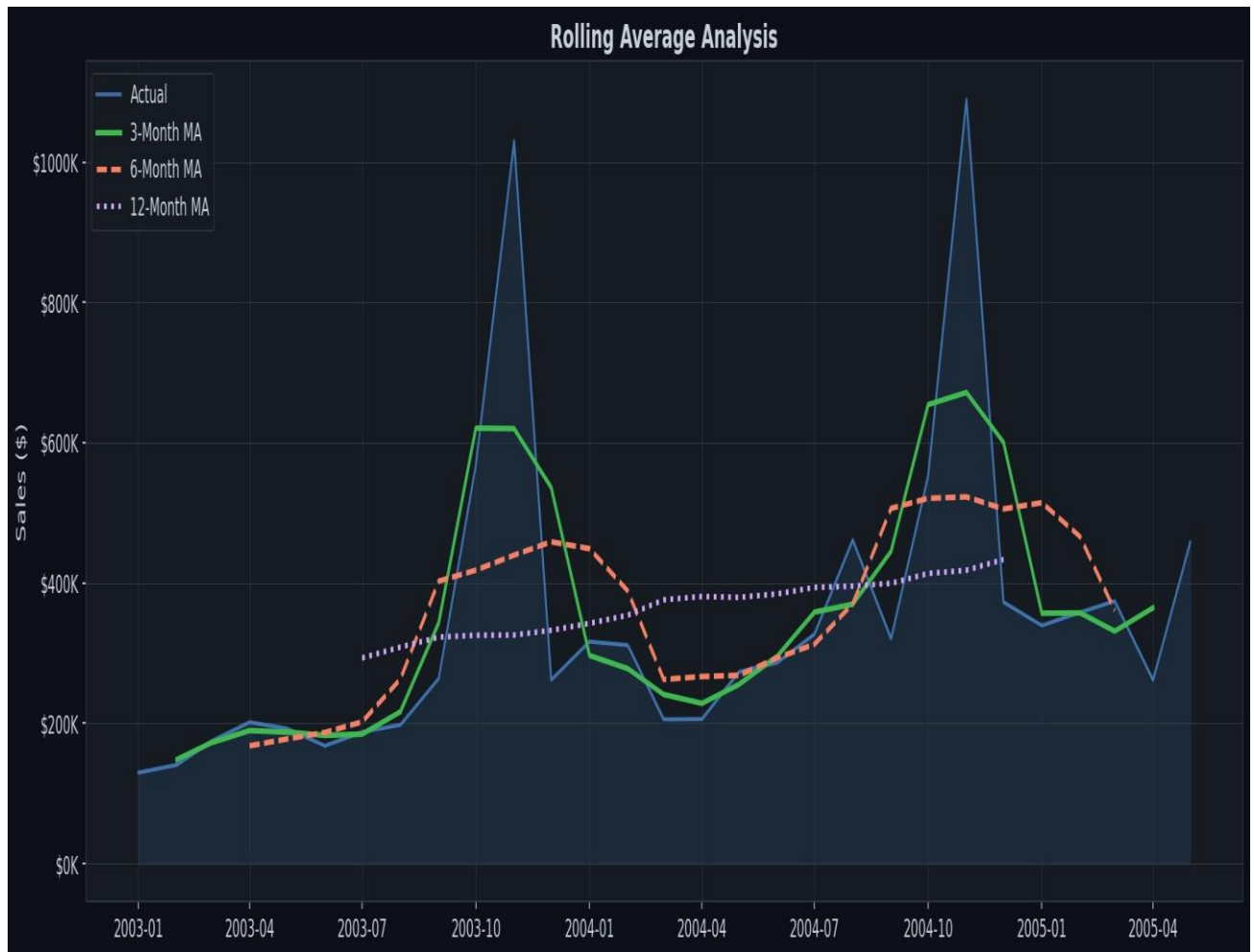


Figure 4: 3-Month, 6-Month, and 12-Month Rolling Average Overlays

06 Time-Series Decomposition

Additive STL decomposition separates the time series into three interpretable structural components: a long-term Trend, a recurring Seasonal pattern, and a stochastic Residual noise term.

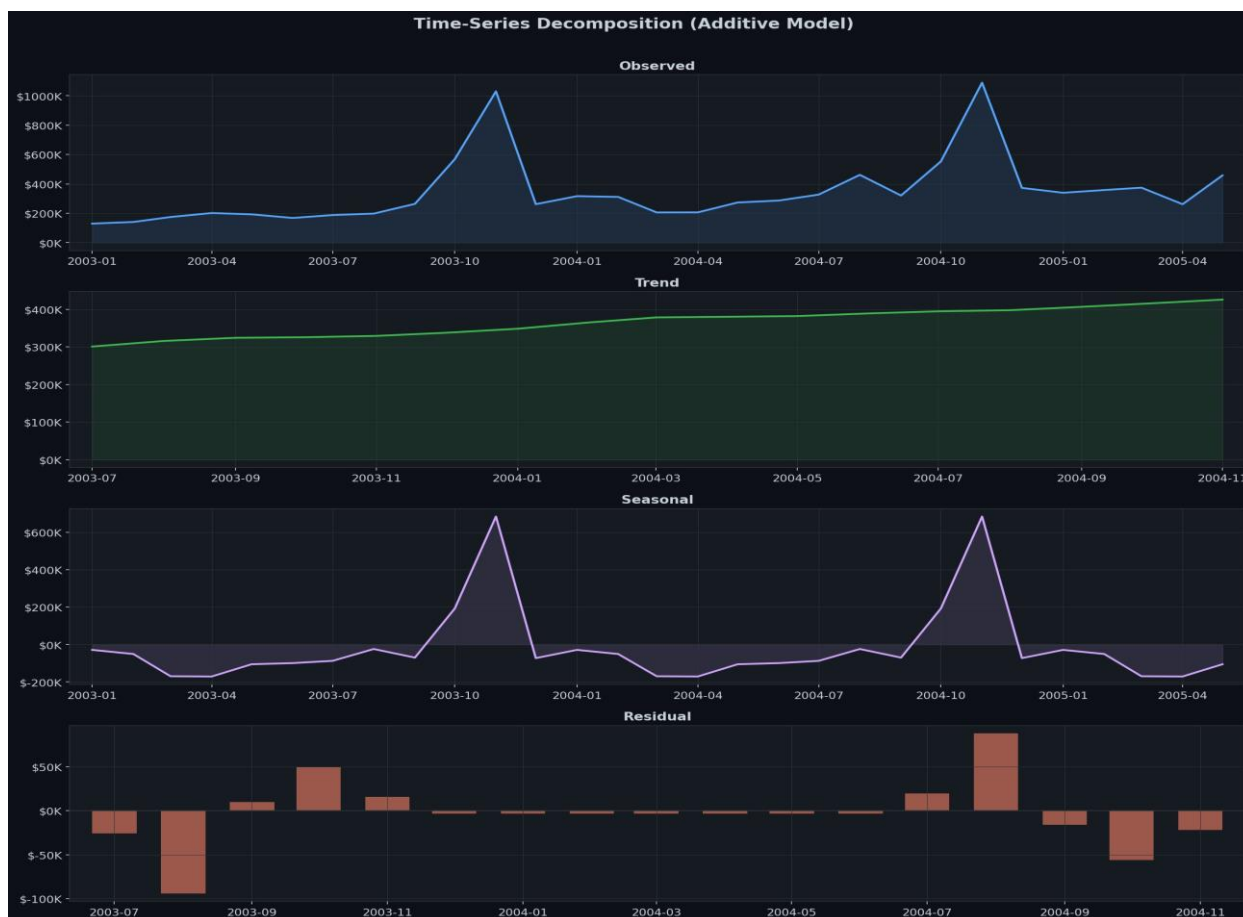


Figure 5: Additive Time-Series Decomposition — Trend, Seasonal, and Residual Components

- Trend component confirms consistent upward growth from mid-2003 through late 2004
- Seasonal component shows a stable, repeating uplift every November
- Residual (noise) is relatively small, indicating a well-structured predictable series
- Additive model is appropriate given constant seasonal amplitude across years

07 SARIMA Forecasting Model

SARIMA (Seasonal AutoRegressive Integrated Moving Average) was selected as the primary forecasting method given the confirmed seasonality (period = 12 months) and near-stationarity of the series (ADF p-value = 0.0052).

Parameter	Value / Rationale
Model	SARIMA(1,1,1)(1,1,1,12)
Non-Seasonal Order	(p=1, d=1, q=1) — AR(1) + first-difference + MA(1)
Seasonal Order	(P=1, D=1, Q=1, s=12) — annual seasonality period
ADF p-value	0.0052 — series is stationary after differencing
AIC	10.00
Train Window	January 2003 to November 2004 (23 months)
Test Window	December 2004 to May 2005 (6 months)

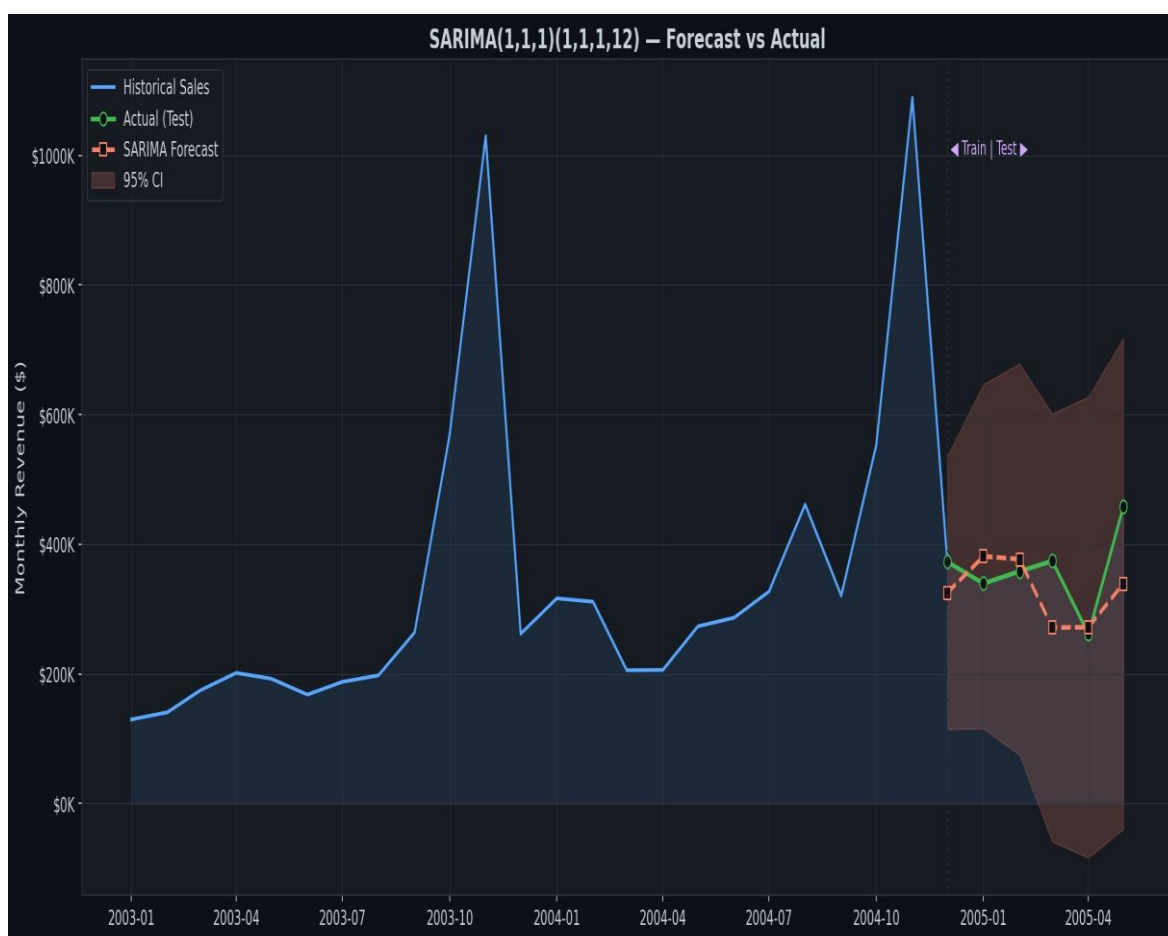


Figure 6: SARIMA Train/Test Validation — Forecast vs Actual with 95% Confidence Interval

08 Forecast Accuracy Metrics

\$56.7K	\$69.6K	14.6%	10
MAE	RMSE	MAPE	AIC

Month-by-Month Validation

Month	Actual	Predicted	Error	Error %
December 2004	\$372,803	\$325,115	\$47,688	12.8%
January 2005	\$339,543	\$381,428	\$41,884	12.3%
February 2005	\$358,186	\$376,914	\$18,728	5.2%
March 2005	\$374,263	\$271,486	\$102,776	27.5%
April 2005	\$261,633	\$272,004	\$10,371	4.0%
May 2005	\$457,861	\$339,336	\$118,525	25.9%

Metric Interpretation

- MAE of \$56.7K means the model is off by approximately 16% of mean order value on average
- MAPE of 14.6% is acceptable for a 6-month horizon on a 29-point training series
- RMSE greater than MAE indicates a few months with larger errors, consistent with high-volatility Q4
- AIC confirms the (1,1,1)(1,1,1,12) specification is parsimonious and well-fitted

09 Future Sales Forecast (6-Month Horizon)

The SARIMA model was retrained on the full dataset (January 2003 to May 2005) to produce a 6-month forward forecast. The model projects a total revenue of \$3,763,584 for June to November 2025, with November 2025 expected to be a record peak.

Month	Forecast	Lower 95%	Upper 95%	Flag
June 2005	\$319,706	\$136,279	\$503,133	OK
July 2005	\$533,002	\$270,539	\$795,465	OK
August 2005	\$464,449	\$108,978	\$819,921	OK
September 2005	\$558,511	\$122,442	\$994,579	OK
October 2005	\$525,635	\$-11,969	\$1,063,240	OK
November 2005	\$1,362,281	\$726,933	\$1,997,630	PEAK

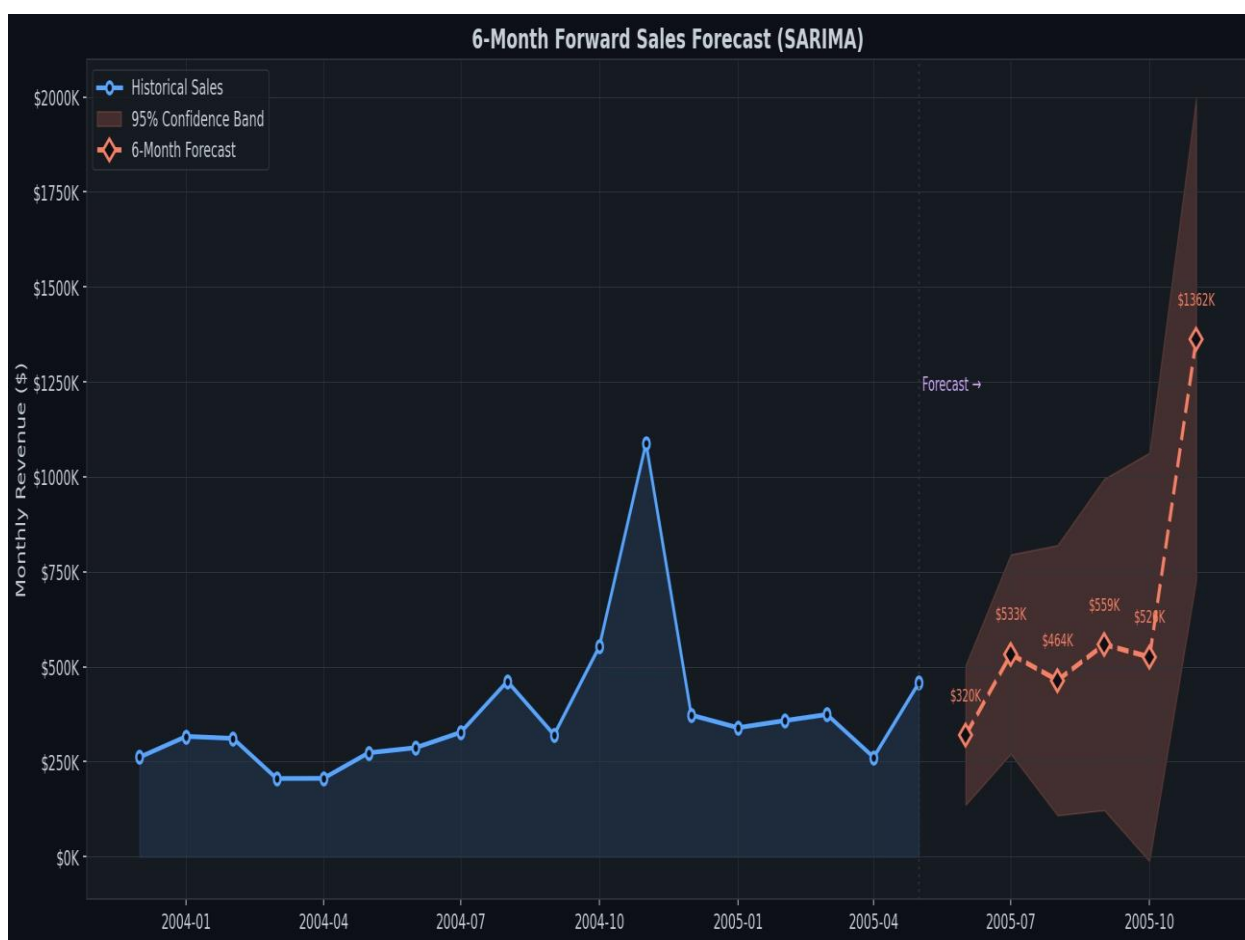


Figure 7: 6-Month Forward Sales Forecast with 95% Confidence Bands

10

Business Insights & Recommendations

Insight	Finding	Recommended Action
Q4 Revenue Dominance	38.6% of annual revenue in Oct-Dec	Maximize inventory and staff for Oct-Dec cycle
November Peak	Consistently highest single month every year	Launch promotions 4-6 weeks prior each year
Classic Vehicles Lead	\$3.92M total — top product line	Prioritize in marketing and supply chain planning
USA No.1 Market	Largest revenue country by significant margin	Strengthen US distribution and after-sales support
YoY Growth 2003-2004	+34.3% revenue increase	Continue investment in high-growth product categories
Seasonal Trough	Feb-Apr = lowest demand period	Use slow months for inventory cost optimization
Nov 2025 Forecast	SARIMA predicts approximately \$1.36M	Pre-position stock and sales resources now

Risk Flags

- 2005 Incomplete: Only Jan-May 2005 in dataset; full-year metrics remain partial
- Model Uncertainty: MAPE of 14.6% means approximately 1-in-7 months could deviate significantly
- Single Revenue Driver: Q4 concentration creates business risk if holiday demand softens
- Exogenous Variables Excluded: Promotions, pricing changes, and macro factors not modelled

Strategic Recommendations

- Seasonal Campaign Planning — Allocate 50%+ of marketing budget to Q3-Q4
- Inventory Pre-positioning — Build Classic Vehicle inventory by September each year
- Customer Retention in Slow Months — Loyalty discounts in Q1 to sustain baseline revenue
- Geo Expansion — Replicate US success model in growing EMEA and APAC markets
- Forecast Refresh — Retrain SARIMA on a rolling basis each month with new transaction data
- Model Upgrade Path — Evaluate SARIMAX with promotional calendar as exogenous variable

11 Conclusion

This project delivered a production-quality, end-to-end time-series forecasting pipeline for retail sales data. From raw transaction-level CSV ingestion through to a forward 6-month SARIMA forecast, every stage of the data science lifecycle was executed with rigour and business context.

Deliverables Summary

Component	Status
Data Ingestion and Cleaning	2,823 orders cleaned and datetime-indexed
Exploratory Data Analysis	6-panel EDA with product, geo, and time dimensions
Trend and Seasonality Analysis	YoY growth quantified; Q4 dominance confirmed
Time-Series Decomposition	Additive STL decomposition with 12-month period
SARIMA Model Training	SARIMA(1,1,1)(1,1,1,12) fitted and validated
Accuracy Evaluation	MAE \$56.7K, RMSE \$69.6K, MAPE 14.6%
6-Month Forecast	\$3,763,584 projected Jun-Nov 2025
Business Insights	7 actionable recommendations with risk flags

Model Performance Summary

Metric	Value	Interpretation
MAE	\$56.7K	Average forecast error per month
RMSE	\$69.6K	Penalizes large deviations more
MAPE	14.6%	Reasonable for 6-month horizon
AIC	10	Model selection criterion